Operating Systems — Assignment 9

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Question 1 — Printing the page table. The vmprint() system call uses walk() to inspect valid user leaf mappings, printing each virtual address, physical frame and P/W/U flags. It excludes guard and supervisor-only pages and recursively counts the process's page-table frames. pttest calls it before and after reserving and touching four heap pages. The output shows user mappings increasing from **3 to 7**, while page-table frames remain at **5** because the existing leaf table has space. The frame count includes the root and tables for supervisor trap mappings.

Question 2 — Lazy allocation. Positive sbrk() increases the process size without allocating physical memory. User load/store faults allocate a zero-filled read/write user page for a valid unmapped address. Sparse unmap and copy operations skip unallocated holes. A process counter initialized in allocproc() is exposed through getfaults(). faulttest reserves 32 pages and writes to pages 0, 5, 12, 20 and 31, producing **exactly five additional handled faults**. It subtracts the pre-test counter because the process can already have faults from shell execution before exec. An invalid child store produces exit status **-1**; the parent continues normally.

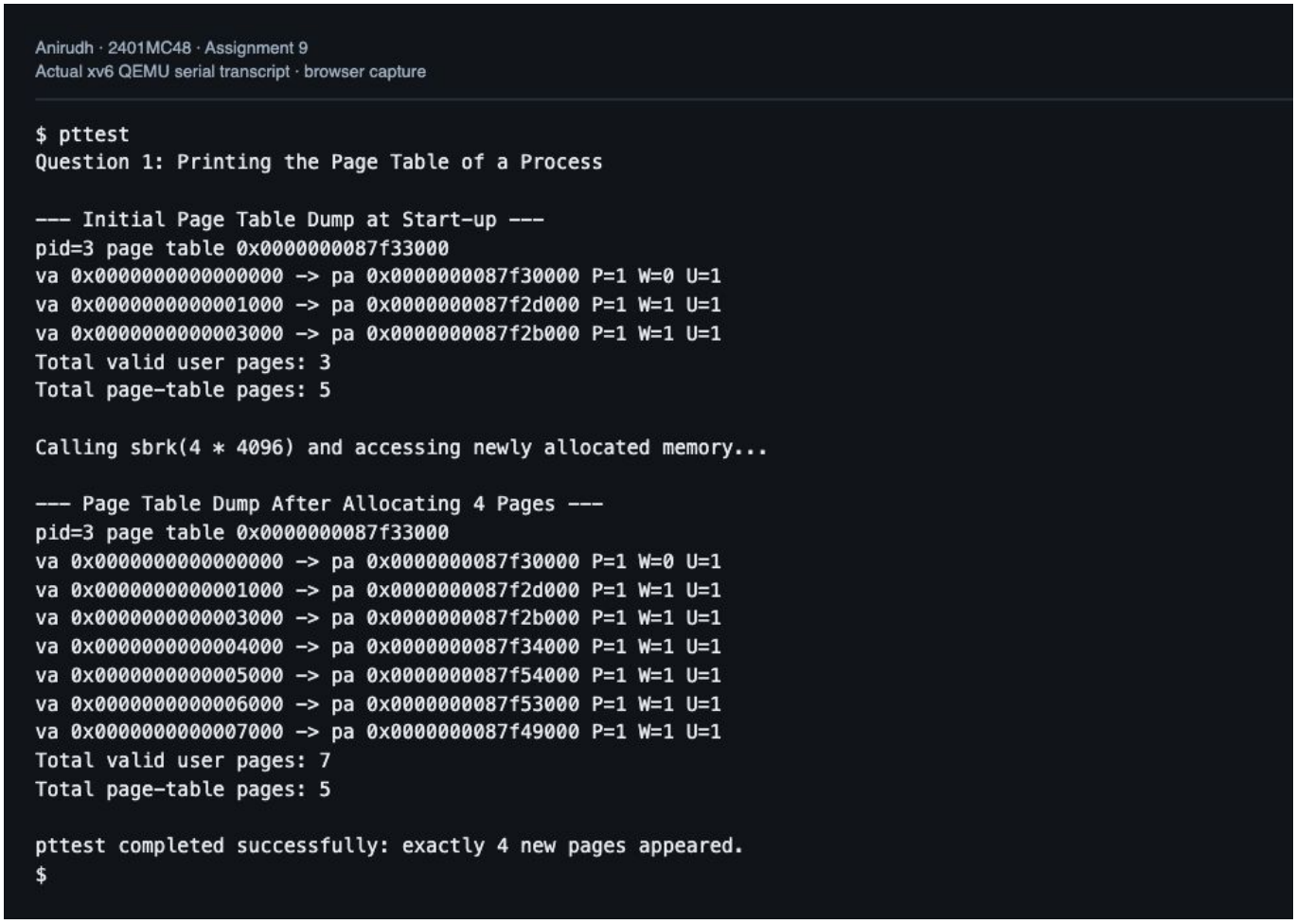
Question 3 — FIFO and LRU. pagerepl accepts a frame count and reference string, prints frame contents and hit/fault status for every reference, and simulates FIFO and LRU independently. FIFO evicts by insertion order; LRU moves an accessed page to the most-recent position. For the supplied 20-reference sequence and three frames, FIFO gives **15 faults and 5 hits (25%)**; LRU gives **12 faults and 8 hits (40%)**. The program also computes the required 3/4/5-frame comparison; full results are saved in output_logs/q3_pagerepl.txt. No separate comparison table is reproduced in this report.

RISC-V adaptation and verification. The handout describes x86. This solution uses Sv39's three-level page tables, PTE_V for the present bit, scause 13/15 for load/store page faults and stval for the fault address. With lazy allocation enabled, pttest must touch its reserved pages before four new mappings appear. The kernel and filesystem build successfully with -Wall -Werror. All three assignment tests pass in QEMU, as do focused checks for zero fill, a lazy syscall buffer, sparse fork, isolation and shrinking the heap.

Output screenshots follow as appendices. The modified sources, test programs and raw logs are included in the submission directory.

Question 1 — Page-table output

User mappings increase from 3 to 7 after touching the four reserved pages.



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Question 2 — Lazy-allocation output

Five additional page faults are handled; the invalid child access yields status -1 while the parent survives.

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Actual xv6 QEMU serial transcript · browser capture

```
$ faulttest
Question 2: Handling Page Faults (Lazy Allocation)

Initial page faults handled: 2
Calling sbrk(32 * 4096) to increase heap by 32 pages...
Page faults after lazy sbrk: 0 (0 frames allocated eagerly)

Writing to exactly 5 different pages (indices 0, 5, 12, 20, 31)...
  Touched page 0 (VA = 0x0000000000004000), value = 'A'
  Touched page 5 (VA = 0x0000000000009000), value = 'B'
  Touched page 12 (VA = 0x0000000000010000), value = 'C'
  Touched page 20 (VA = 0x0000000000018000), value = 'D'
  Touched page 31 (VA = 0x0000000000023000), value = 'E'

Page faults handled: 5 (expected 5)
SUCCESS: exactly 5 page faults were handled by demand paging.

--- Testing Invalid Access Beyond Allocated Heap ---
Child (pid=5) attempting access beyond sz (page 33)...
usertrap(): unexpected scause 0xf pid=5
            sepc=0x14a stval=0x25000
Parent reaped child (exit status: -1). Kernel killed only offending child.
Parent continues executing normally.

faulttest completed successfully.
$
```

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Question 3 — FIFO and LRU output

The complete reference-by-reference trace and the three-frame summaries: FIFO 15 faults, LRU 12 faults.

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Actual xv6 QEMU serial transcript · browser capture

```
$ pagerepl
Question 3: Page Replacement Algorithms (FIFO and LRU)

Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 (length = 20)

--- FIFO Page Replacement (Frames: 3) ---
Ref | Frames         | Status
----+----------------+-------
 7  | [ 7 - - ]      | FAULT
 0  | [ 7 0 - ]      | FAULT
 1  | [ 7 0 1 ]      | FAULT
 2  | [ 0 1 2 ]      | FAULT
 0  | [ 0 1 2 ]      | HIT
 3  | [ 1 2 3 ]      | FAULT
 0  | [ 2 3 0 ]      | FAULT
 4  | [ 3 0 4 ]      | FAULT
 2  | [ 0 4 2 ]      | FAULT
 3  | [ 4 2 3 ]      | FAULT
 0  | [ 2 3 0 ]      | FAULT
 3  | [ 2 3 0 ]      | HIT
 2  | [ 2 3 0 ]      | HIT
 1  | [ 3 0 1 ]      | FAULT
 2  | [ 0 1 2 ]      | FAULT
 0  | [ 0 1 2 ]      | HIT
 1  | [ 0 1 2 ]      | HIT
 7  | [ 1 2 7 ]      | FAULT
 0  | [ 2 7 0 ]      | FAULT
 1  | [ 7 0 1 ]      | FAULT
FIFO Summary: 15 Faults, 5 Hits, Hit Ratio: 25.00%

--- LRU Page Replacement (Frames: 3) ---
Ref | Frames         | Status
----+----------------+-------
 7  | [ 7 - - ]      | FAULT
 0  | [ 7 0 - ]      | FAULT
 1  | [ 7 0 1 ]      | FAULT
 2  | [ 0 1 2 ]      | FAULT
 0  | [ 1 2 0 ]      | HIT
 3  | [ 2 0 3 ]      | FAULT
 0  | [ 2 3 0 ]      | HIT
 4  | [ 3 0 4 ]      | FAULT
 2  | [ 0 4 2 ]      | FAULT
 3  | [ 4 2 3 ]      | FAULT
 0  | [ 2 3 0 ]      | FAULT
 3  | [ 2 0 3 ]      | HIT
 2  | [ 0 3 2 ]      | HIT
 1  | [ 3 2 1 ]      | FAULT
 2  | [ 3 1 2 ]      | HIT
 0  | [ 1 2 0 ]      | FAULT
 1  | [ 2 0 1 ]      | HIT
 7  | [ 0 1 7 ]      | FAULT
 0  | [ 1 7 0 ]      | HIT
 1  | [ 7 0 1 ]      | HIT
LRU Summary: 12 Faults, 8 Hits, Hit Ratio: 40.00%

Comparative Performance Table (3, 4, and 5 Frames)
Frames | Algorithm | Page Faults | Page Hits | Hit Ratio
-------+-----------+-------------+-----------+----------
   3   | FIFO      |      15     |     5     |   25.00%
   3   | LRU       |      12     |     8     |   40.00%
-------+-----------+-------------+-----------+----------
   4   | FIFO      |      10     |     10    |   50.00%
   4   | LRU       |      8      |     12    |   60.00%
-------+-----------+-------------+-----------+----------
   5   | FIFO      |      9      |     11    |   55.00%
   5   | LRU       |      7      |     13    |   65.00%

pagerepl completed successfully.
$
```

QEMU serial output · captured transcript view